

nonzero levels in value space

exact zero

0

$$q_i = 0$$

$2^{\ell_{\min}}$

$2^{\ell_{\max}}$



uniform code steps in $\log_2 x \Rightarrow$ nonuniform levels in value space

$$q_i = 0 \Leftrightarrow x_i = 0$$

AL8: 255 nonzero codes • AL16: 65,535 nonzero codes

each block adapts $[\ell_{\min}, \ell_{\max}]$ to its observed nonzero range